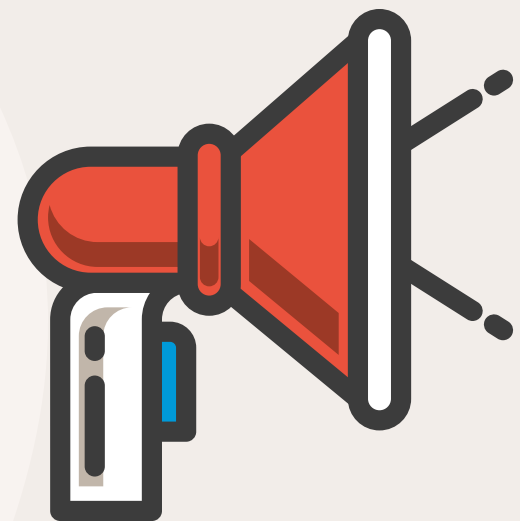




INTERNSHIP PRESENTATION



Student Name: Phoong Guik Ming
Student ID: 2006616
Faculty: FICT





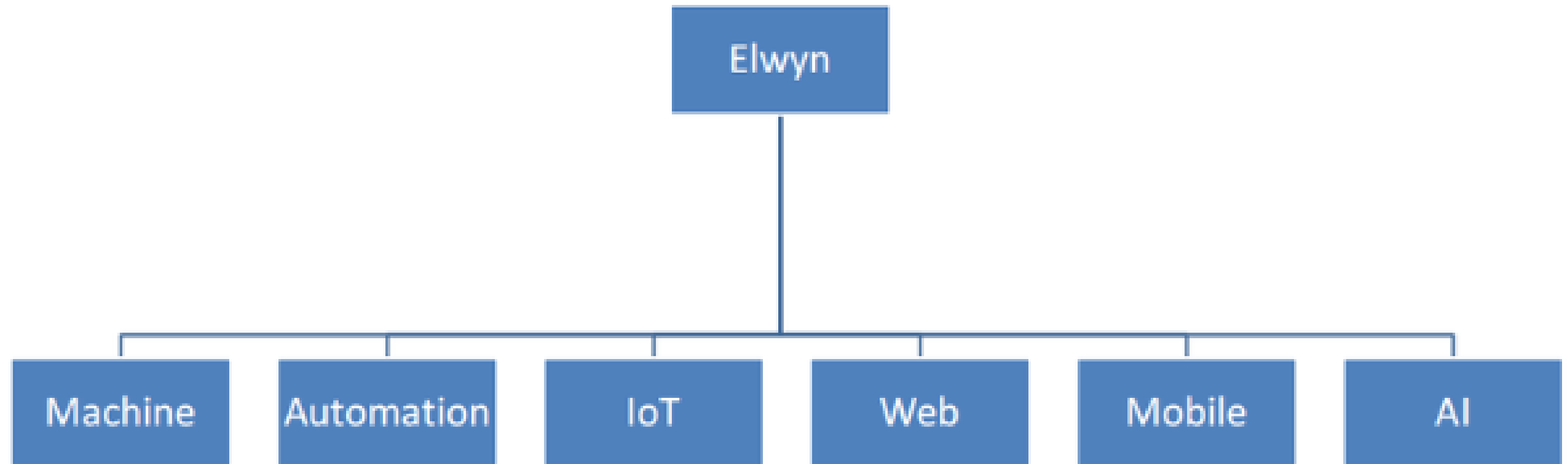
COMPANY BACKGROUND

- Sophic Automation Sdn Bhd was founded in 2007 with the mission of providing the most innovative and effective Industrial Automation Solutions to customers.
- Sophic offer an attractively priced and complete outsourcing mechanism, through which sophic develop, supply, and manage customized automation projects for customers.



JOB FUNCTION

Software Innovation

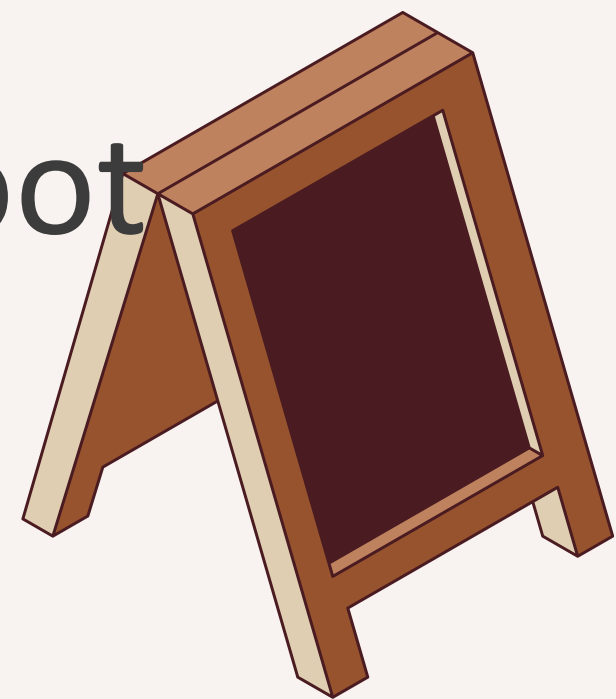


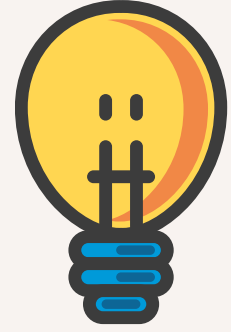


JOB FUNCTION



- **Data Extraction:** Extract data from multiple sources such as textual, web pages, API, formatted report.
- **Data Conversion:** Standardize formats (CSV → JSON) for processing and storage.
- **REST API:** Build Flask-based APIs for real-time queries and results delivery.
- **Data Analysis:** Analyze fluctuations, identify root causes, and develop actionable insights.



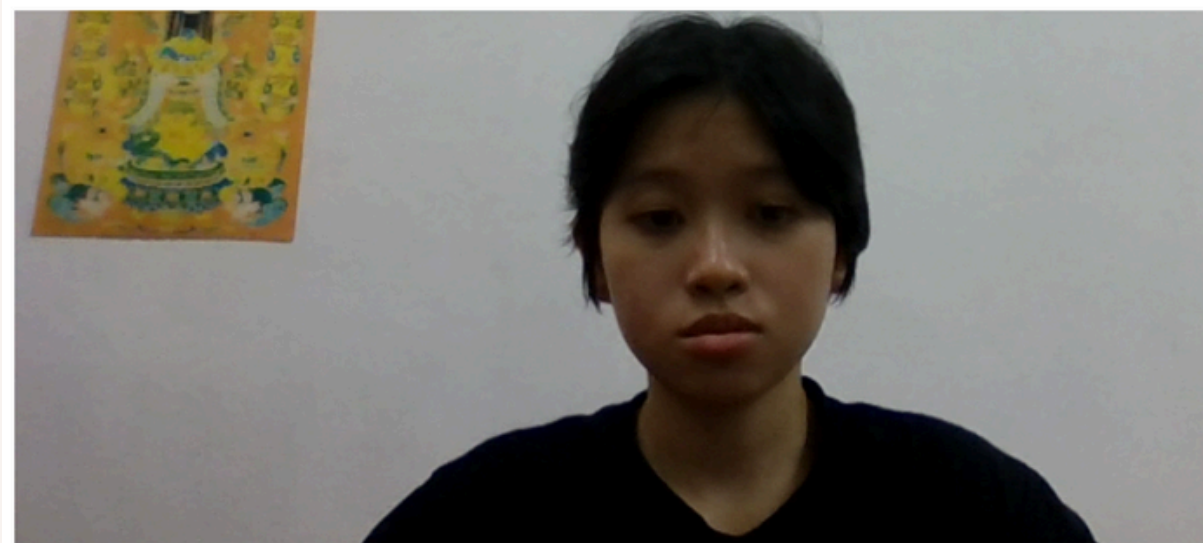


WHAT HAS BEEN DONE BY INTERN DURING THE INDUSTRIAL TRAINING

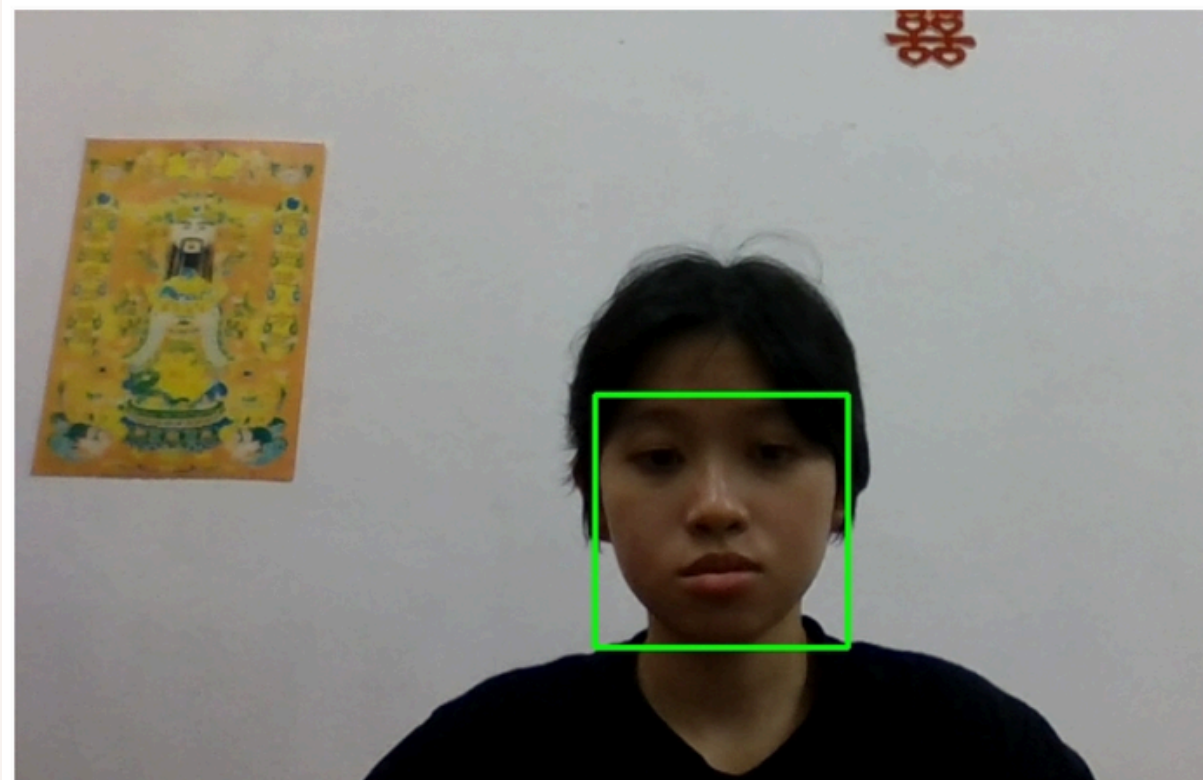


STREAMLIT

Objectives: Create user profiles by recognizing faces



x Clear photo



Captured Image with Face Detection

Welcome, yueming! Your profile has been created successfully!

User Information

Name: yueming

Age: 33

Email: asdf@gmail.com

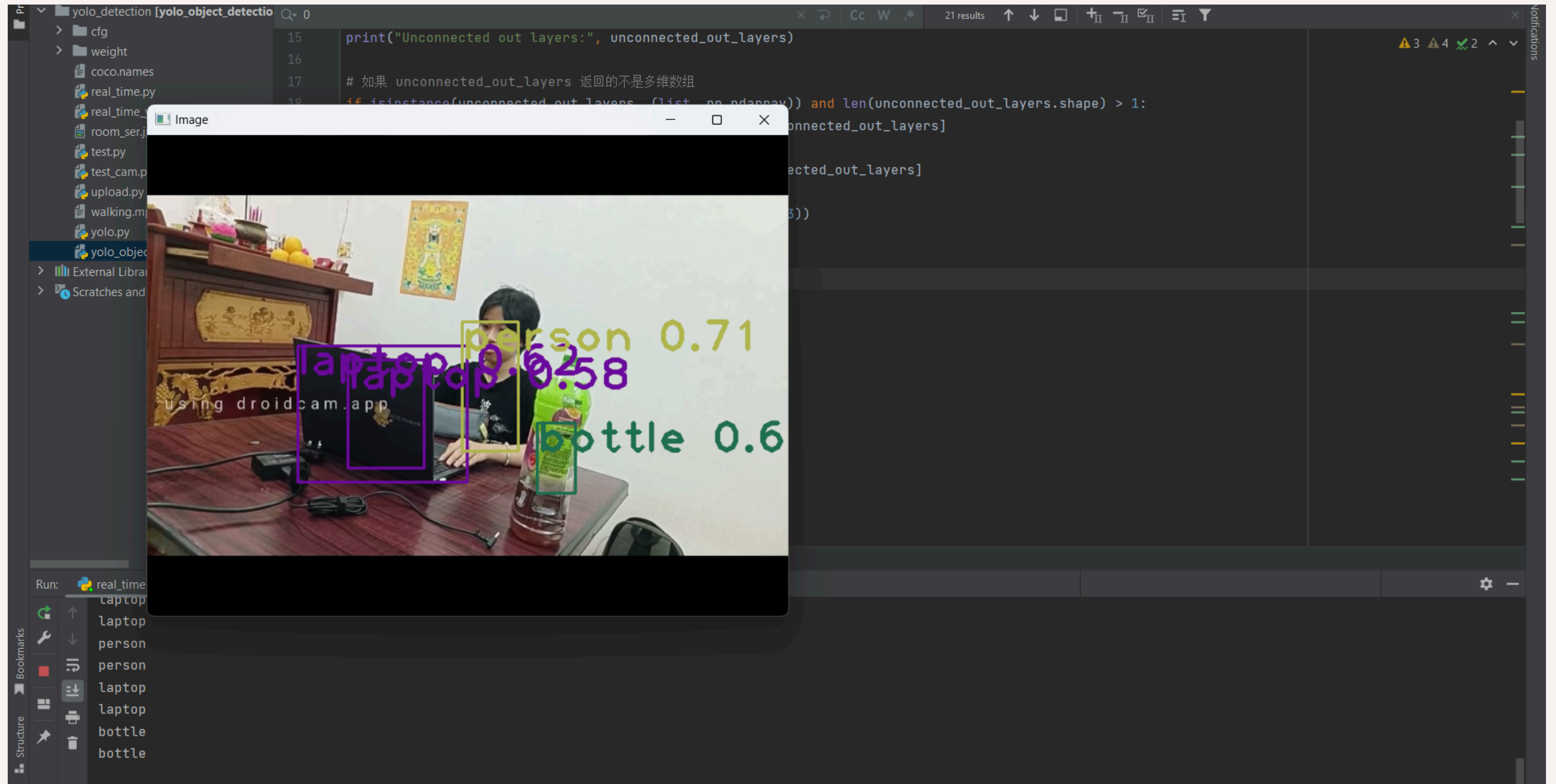
Registered: 2024-11-06



Profile Photo - yueming

Log Out

OBJECT-DETECTION SYSTEM



TRANSFER CSV TO JSON

```
scrap1

ess_out
ng_scratch
ages
s
sure_scr
_spot_dot
r_stain
ing_scr
ss_scratches
at_hole
h_measure_scr
airline
llanous
stream
at_edge
s
ting
s_bubble
r
_contamin
ess_out
at_hole
_spot_dot
ages
stream
ng_scratch
airline
e bubble

TERMINAL PORTS AZURE

parkraw2_timpiaria.csv
ve\Desktop\sophic\scrap1> & C:/Users/guikm_awtx1ty/OneDrive/Desktop/sophic/scrap1/spark3wei.py
parkraw2_timpiaria.csv
ve\Desktop\sophic\scrap1> & C:/Users/guikm_awtx1ty/OneDrive/Desktop/sophic/scrap1/merge.py
parkraw2_timpiaria.csv'
ve\Desktop\sophic\scrap1>
ve\Desktop\sophic\scrap1>

ve\Desktop\sophic\scrap1>
```

```
scrap1

ed.csv
1 > [ ] defects

ess": "After Etching CRVI",
ects": [
'stain_contamin",
'flatness_out",
'matting_scratch",
'breakages",
'others",
'qc_mesure_scr",
'white_spot_dot",
'powder_stain",
'handling_scr",
'scratches",
'chip_at_hole",
'polish_measure_scr",
'scr_hairline",
'miscellaneous",
'lava_stream",
'chip_at_edge",
'cracks",
'unmatting",
'inclus_bubble",
'powder"

CONSOLE TERMINAL PORTS AZURE

parkraw2_timpiaria.csv
ve\Desktop\sophic\scrap1> & C:/Users/guikm_awtx1ty/OneDrive/Desktop/sophic/scrap1/aiPy
sophic/scrap1/spark3wei.py
parkraw2_timpiaria.csv
ve\Desktop\sophic\scrap1> & C:/Users/guikm_awtx1ty/OneDrive/Desktop/sophic/scrap1/aiPy
sophic/scrap1/merge.py
merged_sparkraw2_timpiaria.csv'
ve\Desktop\sophic\scrap1>
ve\Desktop\sophic\scrap1>

ve\Desktop\sophic\scrap1>
```


WEB SCRAPING

Fake Python

realpython.github.io/fake-jobs/

Fake Python

Fake Jobs for Your Web Scraping

Senior Python Developer

Payne, Roberts and Davis

Stewartbury, AA

2021-04-08

Learn

Legal executive

Jackson, Chambers and Levy

Port Ericaburgh, AA

2021-04-08

Learn

Product manager

Ramirez Inc

Run Terminal Help

scrap

tar.py jobs.csv X

jobs.csv > data

```
1 Job Title,Company Name,Location,Date Posted
2 Senior Python Developer,"Payne, Roberts and Davis","Stewartbury, AA",2021-04-08
3 Energy engineer,Vasquez-Davidson,"Christopherville, AA",2021-04-08
4 Legal executive,"Jackson, Chambers and Levy","Port Ericaburgh, AA",2021-04-08
5 Fitness centre manager,Savage-Bradley,"East Seanview, AP",2021-04-08
6 Product manager,Ramirez Inc,"North Jamieview, AP",2021-04-08
7 Medical technical officer,Rogers-Yates,"Davidville, AP",2021-04-08
8 Physiological scientist,Kramer-Klein,"South Christopher, AE",2021-04-08
9 Textile designer,Meyers-Johnson,"Port Jonathan, AE",2021-04-08
10 Television floor manager,Hughes-Williams,"Osbornetown, AE",2021-04-08
11 Waste management officer,"Jones, Williams and Villa","Scotttown, AP",2021-04-08
12 Software Engineer (Python),Garcia PLC,"Ericberg, AE",2021-04-08
13 Interpreter, Gregory and Sons,"Ramireztown, AE",2021-04-08
```

USD/JPY
+0.67%

Search

10:48 PM
12/15/2024

FLASK REST API

The image displays a web browser and a code editor (VS Code) side-by-side, illustrating a Flask REST API running on localhost:5000.

Browser View (Left): The address bar shows `127.0.0.1:5000/api/users/4`. The response is a JSON object:

```
1 {
2   "id": 4,
3   "name": "YouMing",
4   "email": "youming@gmail.com"
5 }
```

VS Code View (Right): The Explorer pane shows the project structure for `FLASKK1`, including files like `api.py`, `api_db.py`, `flask1.py`, `flask2.py`, `flaskAPI.py`, `requirements.txt`, `server.py`, `testapi.py`, and `weather.py`.

The REST client interface shows a GET request to `http://localhost:5000/api/users/4` with a JSON body:

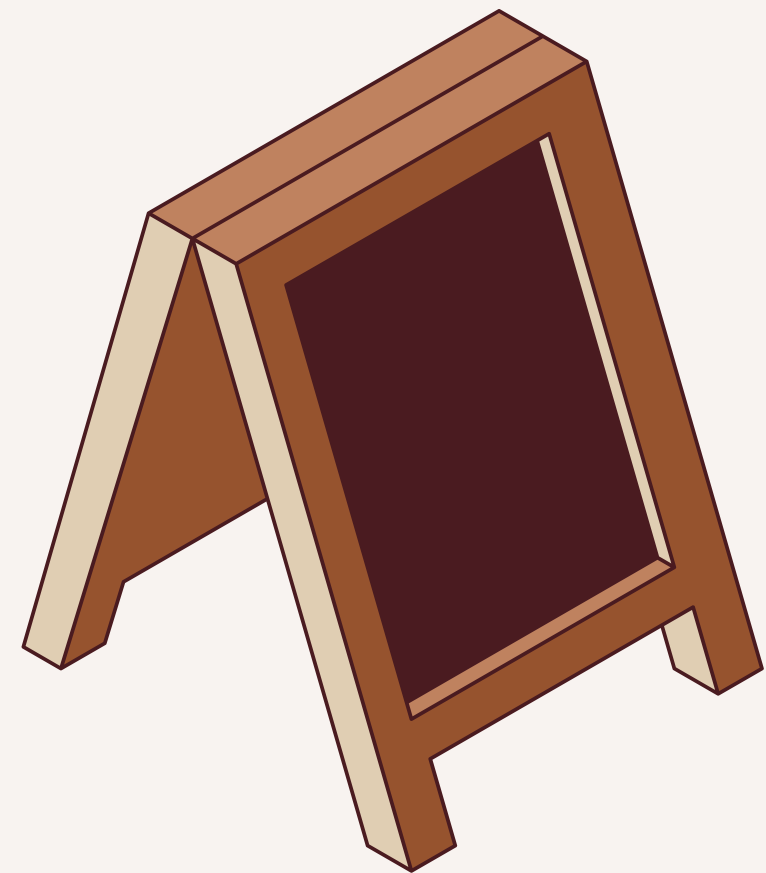
```
1 { "name": "YouMing", "email": "youming@gmail.com" }
```

The status bar indicates **Status: 200 OK**, **Size: 73 Bytes**, and **Time: 5 ms**.

The Terminal pane shows the following log output:

```
"POST /api/users HTTP/1.1" 308 -
127.0.0.1 - - [19/Nov/2024 08:45:21] "POST /api/users/ HTTP/1.1" 201
127.0.0.1 - - [19/Nov/2024 08:45:58] "GET /api/users HTTP/1.1" 308 -
127.0.0.1 - - [19/Nov/2024 08:45:58] "GET /api/users/ HTTP/1.1" 200 -
127.0.0.1 - - [19/Nov/2024 08:46:02] "GET /api/users/4 HTTP/1.1" 200
127.0.0.1 - - [19/Nov/2024 08:46:15] "GET /api/users/4 HTTP/1.1" 200
```

RELATIONSHIP BETWEEN WATER LEVEL OF PUMPHOUSE AND WEATHER



RAW DATA

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	id	battery_vol	latest_data	latest_update																
2	1	3.6	0.413	3/23/2023 17:22																
3	2	3.6	0.413	3/23/2023 17:24																
4	3	3.6	0.413	3/23/2023 17:26																
5	4	3.6	0.413	3/23/2023 17:28																
6	5	3.6	0.413	3/23/2023 17:31																
7	6	3.6	0.413	3/23/2023 17:33																
8	7	3.6	0.413	3/23/2023 17:35																
9	8	3.6	0.413	3/23/2023 17:37																
10	9	3.6	0.412	3/23/2023 17:39																
11	10	3.6	0.412	3/23/2023 17:41																
12	11	3.6	0.413	3/23/2023 17:43																
13	12	3.6	0.412	3/23/2023 17:45																
14	13	3.6	0.412	3/23/2023 17:47																
15	14	3.6	0.412	3/23/2023 17:50																
16	15	3.6	0.412	3/23/2023 17:52																
17	16	3.6	0.412	3/23/2023 17:54																
18	17	3.6	0.412	3/23/2023 17:56																
19	18	3.6	0.412	3/23/2023 17:58																
20	19	3.6	0.412	3/23/2023 18:00																
21	20	3.6	0.412	3/23/2023 18:02																
22	21	3.6	0.412	3/23/2023 18:05																
23	22	3.6	0.412	3/23/2023 18:07																
24	23	3.6	0.412	3/23/2023 18:09																
25	24	3.6	0.412	3/23/2023 18:11																
26	25	3.6	0.412	3/23/2023 18:13																
27	26	3.6	0.412	3/23/2023 18:15																
28	27	3.6	0.412	3/23/2023 18:17																
29	28	3.6	0.412	3/23/2023 18:19																
30	29	3.6	0.412	3/23/2023 18:22																
31	30	3.6	0.412	3/23/2023 18:24																

MERGE DATA WITH WEATHER

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
battery_vol	latest_data	latest_update	temp	precip	sealevelpre	solarradiati	solarenergy	severerisk	icon	description					
3.6	0.413	3/23/2023	83.2	0.034	1008.1	292	25.3	60	rain	Partly cloudy throughout the day with rain in the morning and afternoon.					
3.6	0.408	3/24/2023	82.9	0.017	1008.4	299.3	25.8	60	rain	Partly cloudy throughout the day with afternoon rain.					
3.6	0.402	3/25/2023	83.1	0	1009.1	301.5	25.9	60	partly-cloud	Partly cloudy throughout the day.					
3.6	0.402	3/26/2023	83.6	0.003	1008.3	300.1	25.8	60	rain	Partly cloudy throughout the day with morning rain.					
3.6	0.402	3/27/2023	82.6	0.175	1008.1	243.7	21.1	60	rain	Partly cloudy throughout the day with rain.					
3.6	0.406	3/28/2023	82.3	0	1009.1	295.2	25.5	60	partly-cloud	Partly cloudy throughout the day.					
3.6	0.407	3/29/2023	83.2	0	1009.5	303.2	26.2	60	partly-cloud	Partly cloudy throughout the day.					
3.6	0.414	3/30/2023	83.8	0.01	1008.5	313.6	27	30	rain	Partly cloudy throughout the day with morning rain.					

Addresses, partial addresses or lat,lon

History or forecast data

Data units

Query options

Data sections Weather elements Degree days Wind & solar Agriculture Weather stations



API



Grid



Chart



JSON



CSV

Weather Data

Additional Data

Data Details



Daily



Hourly



Current



Events



Alerts



Info



Stations

Questions about the data?

Download

Available weather data for Butterworth, Pulau Pinang, Malaysia. These results are filtered by your query options.

datetime	tempmax	tempmin	temp	feelslikemax	feelslikemin	feelslike	dew	humidity	precip	precipprob	precipcover	preciptype	snow	snowdepth	windgust	windspeed	winddir	pr
2024-12-15	89.4	77	82.4	98.1	77	87.6	74.4	77.4	0.084	100	37.5	rain	0	0	6.9	7.8	343.6	10
2024-12-16	85.7	74	78.7	96	74	81.8	74.8	88.3	0.263	96.8	54.17	rain	0	0	6.9	4.9	38.1	10
2024-12-17	80	73.7	76.2	80	73.7	76.2	73.6	91.8	0.26	90.3	41.67	rain	0	0	8.7	5.1	51	10
2024-12-18	80.1	72.1	75.3	84.9	72.1	75.7	72.4	90.8	0.253	100	54.17	rain	0	0	8.5	5.1	98.2	10
2024-12-19	75.6	71.7	73.6	75.6	71.7	73.6	71.7	93.9	0.45	87.1	58.33	rain	0	0	12.1	4.5	61.6	10
2024-12-20	80.5	72.4	76	85.5	72.4	76.4	72.3	88.6	0.166	93.5	20.83	rain	0	0	3.6	3.8	89.9	10
2024-12-21	86.3	72.4	77.6	94.2	72.4	79.9	72.8	85.9	0.217	96.8	20.83	rain	0	0	5.4	4	80.8	10
2024-12-22	85.4	75.4	79.5	95.1	75.4	79.7	75.4	85.4	0.288	95.9	25.83	rain	0	0	5.4	4.5	98.7	10
2024-12-23	86.4	75.2	79.8	95.3	75.2	79.7	75.3	84.3	0.288	95.9	25.83	rain	0	0	8.1	7.6	23.3	10
2024-12-24	86.8	75	79.3	95	75	79.4	80.3	79.7	0	79.2	0	rain	0	0	10.7	9.6	27.9	10

×

pumphous

×

Untitled3

×

raw.git

3888/notebooks/OneDrive/Desktop/sophic/pumphouse/

book 历史经典文丛 (套... translate - Google...

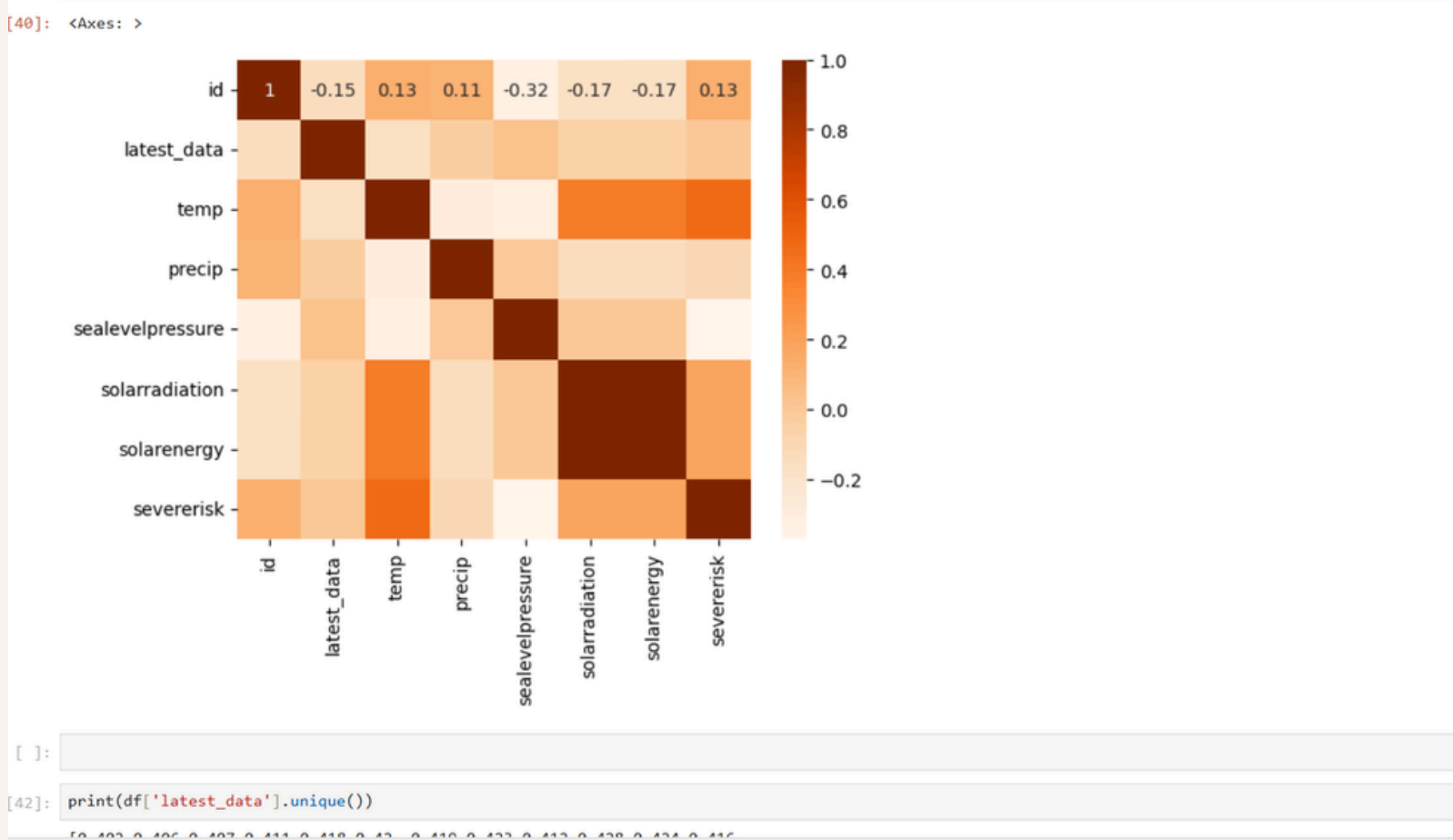
jupyter pumphouse1 (1) Last Checkpoint: 4 da

Edit View Run Kernel Settings Help

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JupyterLab Python 3 (ipykerr)

```
[40]: #create a heatmap for visualizing the correlation matrix
sns.heatmap(corr_matrix, annot=True, cmap="Oranges")
```



+ 🔍 📄 📁 ▶ ⏮ ⏪ ⏩ ⏭ Code ▾

JupyterLab Python 3 (ipykernel)

```
[8]: df.describe()
```

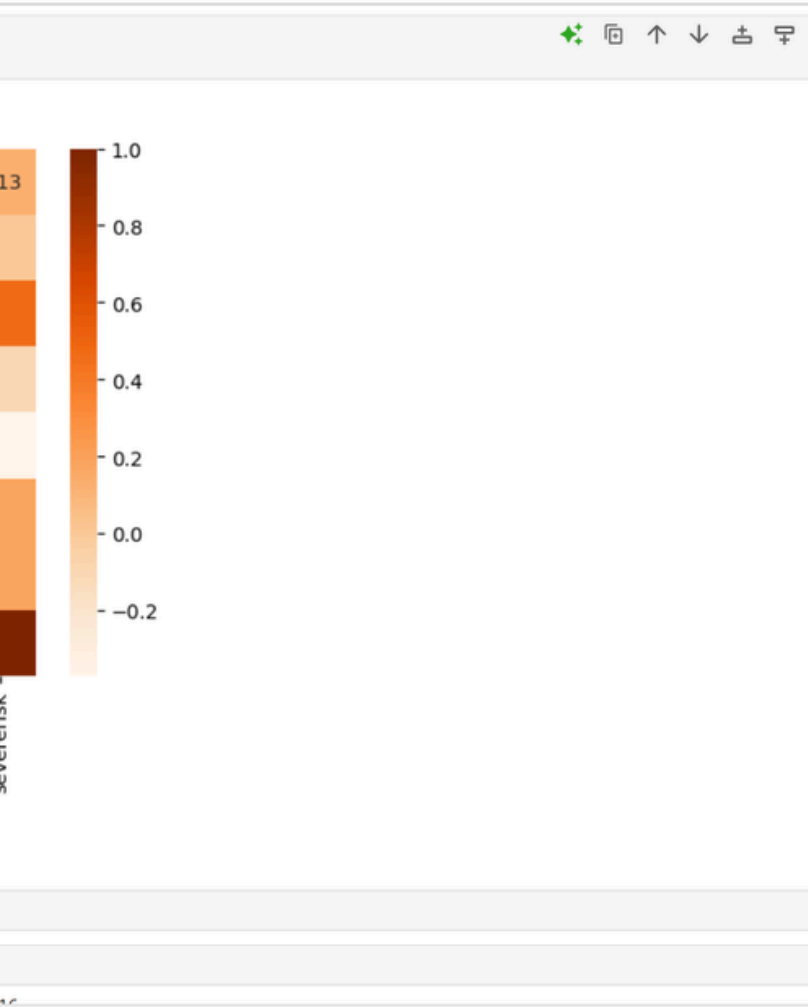
	battery_voltage	latest_data	temp	precip	sealevelpressure	solarradiation	solarenergy	severerisk
count	5.020000e+02	502.000000	502.000000	502.000000	502.000000	502.000000	502.000000	502.000000
mean	3.600000e+00	0.236145	83.649402	0.083217	1009.355976	234.531275	20.251793	40.866534
std	3.022819e-14	0.636087	2.009704	0.199908	1.406612	64.045603	5.529780	18.147657
min	3.600000e+00	0.000000	77.900000	0.000000	1005.500000	23.900000	2.200000	10.000000
25%	3.600000e+00	0.019000	82.200000	0.000000	1008.400000	211.525000	18.200000	30.000000
50%	3.600000e+00	0.021000	83.600000	0.000000	1009.400000	254.550000	22.000000	30.000000
75%	3.600000e+00	0.073000	85.000000	0.038000	1010.200000	279.800000	24.200000	60.000000
max	3.600000e+00	9.999000	88.300000	1.472000	1013.700000	313.600000	27.100000	100.000000

```
[14]: df.describe().T
```

	count	mean	std	min	25%	50%	75%	max
battery_voltage	502.0	3.600000	3.022819e-14	3.6	3.600	3.600	3.600	3.600
latest_data	502.0	0.236145	6.360868e-01	0.0	0.019	0.021	0.073	9.999
temp	502.0	83.649402	2.009704e+00	77.9	82.200	83.600	85.000	88.300
precip	502.0	0.083217	1.999084e-01	0.0	0.000	0.000	0.038	1.472
sealevelpressure	502.0	1009.355976	1.406612e+00	1005.5	1008.400	1009.400	1010.200	1013.700

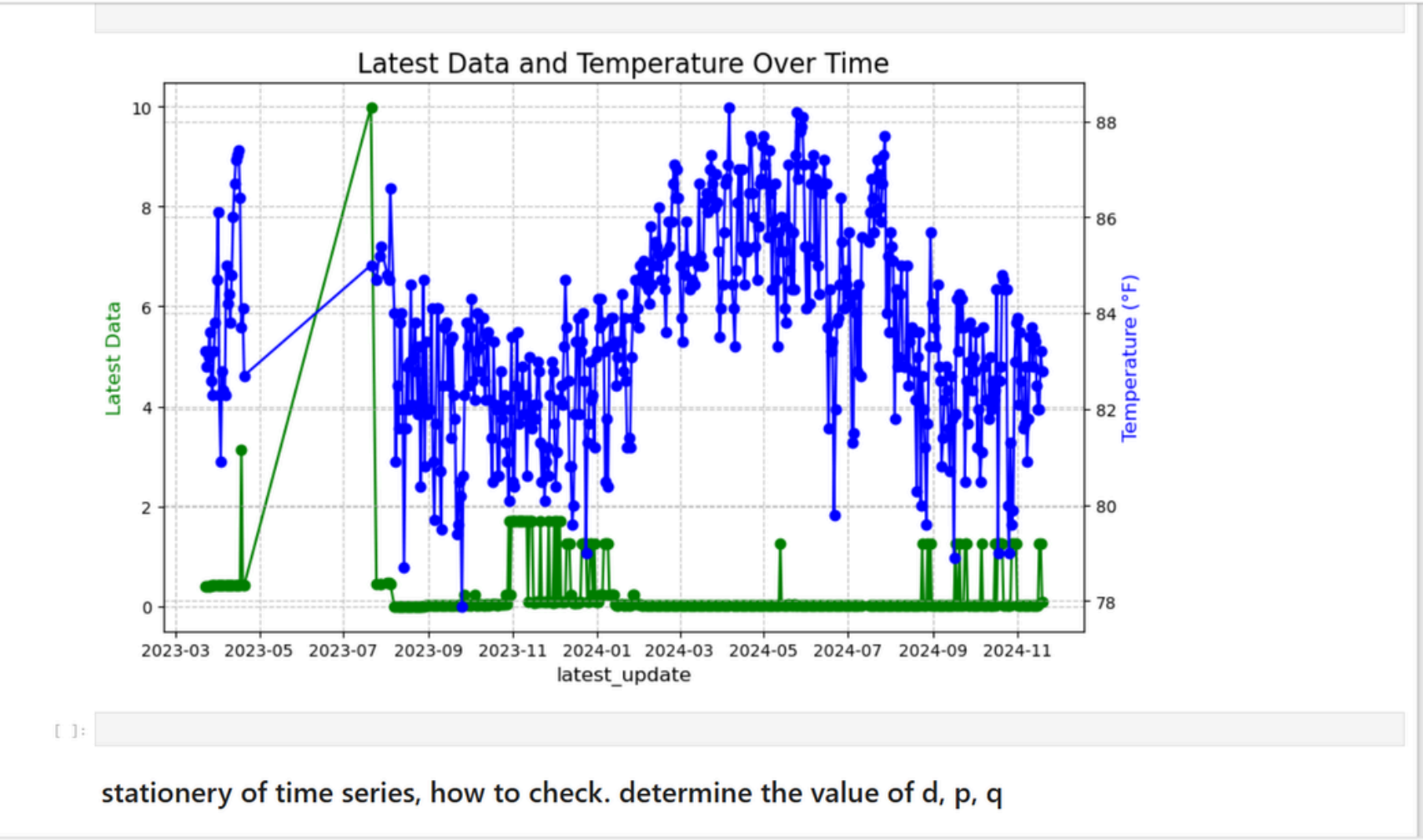
+ 🔍 📄 📁 ▶ ⏮ ⏪ ⏩ ⏭ Code ▾

JupyterLab Python 3 (ipykerr)



+ 🔍 📄 📁 ▶ ⏮ ⏪ ⏩ ⏭ Code ▾

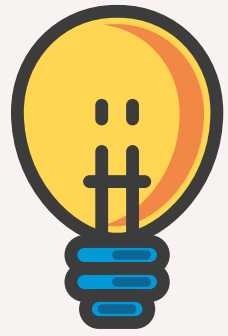
JupyterLab Python 3 (ipykernel)



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019春... 2019東禪佛學院【...

Trusted



ARIMA TIME SERIES MODEL

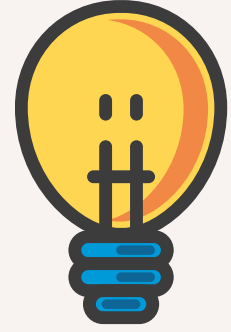
Accuracy Metrics for Time Series Forecast



```
# Define forecast accuracy function
def forecast_accuracy(forecast, actual):
    forecast = np.array(forecast)
    actual = np.array(actual)
    mape = np.mean(np.abs(forecast - actual) / np.abs(actual)) # MAPE
    me = np.mean(forecast - actual) # ME
    mae = np.mean(np.abs(forecast - actual)) # MAE
    mpe = np.mean((forecast - actual) / actual) # MPE
    rmse = np.mean((forecast - actual) ** 2) ** .5 # RMSE
    corr = np.corrcoef(forecast, actual)[0, 1] # corr
    mins = np.amin(np.hstack([forecast[:, None], actual[:, None]]), axis=1)
    maxs = np.amax(np.hstack([forecast[:, None], actual[:, None]]), axis=1)
    minmax = 1 - np.mean(mins / maxs) # minmax
    acf1 = acf(forecast - actual)[1] # ACF1
    return {'mape': mape, 'me': me, 'mae': mae, 'mpe': mpe, 'rmse': rmse, 'acf1': acf1, 'corr': corr, 'minmax': minmax}

# Calculate forecast accuracy
arima_accuracy = forecast_accuracy(fc_series, test_single)
print(f'ARIMA accuracy: {arima_accuracy}')
```

```
ARIMA accuracy: {'mape': 1.9150531166010705, 'me': -0.30710163575657917, 'mae': 0.3117743198056307, 'mpe': 1.5884273183596416, 'rmse': 0.8124350419840117, 'acf1': 0.5266005260548933, 'corr': 0.019081415243055407, 'minmax': -1.6021873948813394}
```



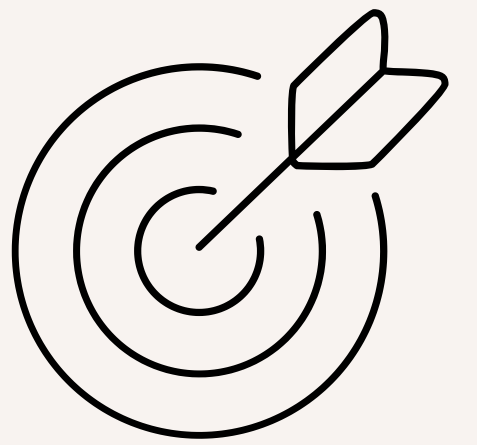
WHAT HAS THE INTERN LEARNT IN THE INDUSTRIAL TRAINING

(THINGS THAT INTERN HAS NOT LEARNT IN UTAR)





SOFT SKILLS



- **Proactive Attitude:** Learned the importance of asking questions when encountering challenges or uncertainties.
- **Independent Research Skills:** Developed the ability to effectively utilize online resources to solve problems and gain new knowledge.
- **Time Management:** Improved the skill of organizing and prioritizing tasks to meet deadlines efficiently.

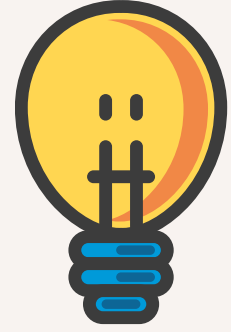


HARD SKILLS



- Object Detection
- Web Scraping
- Flask REST API
- Water Level Analysis



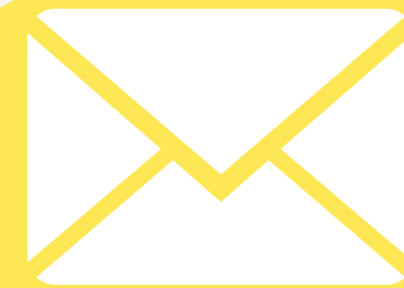


WHAT INTERN HAS APPLIED IN THE INDUSTRIAL TRAINING

(THINGS THAT INTERN HAS LEARNT FROM UTAR)



- 
- 
- Jupyter (Python)
 - Visual Studio Code (Python)





Thank You

